

NO 4D X

Why There Is no 4-Dimensional Cross Product

Axioms for the Cross Product

For $a, b, c \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$

- $b \times a = -a \times b$ (skew symmetric)
- $a \times (b + c) = (a \times b) + (a \times c)$
 $(\lambda a) \times b = \lambda(a \times b)$ (bilinear)
- $a \cdot (a \times b) = 0$ (produces an orthogonal vector)
- $|a \times b|^2 = |a|^2 |b|^2 - (a \cdot b)^2$ (the cross product of orthogonal unit vectors is a unit vector)

Question: What are the possible values of n ?

(from "The Scarcity of Cross Products on Euclidean Spaces

B. Walsh, Amer. Math. Monthly 1967)

$$\boxed{n=1} \quad \mathbb{R} = \text{span} \{e_1\}$$

$$\begin{aligned} a, b \in \mathbb{R} &\Rightarrow a = \lambda_1 e_1 & a \times b &= (\lambda_1 e_1) \times (\lambda_2 e_1) \\ & b = \lambda_2 e_1 & &= \lambda_1 \lambda_2 (e_1 \times e_1) = 0 \\ & & &= - (e_1 \times e_1) \Rightarrow = 0 \end{aligned}$$

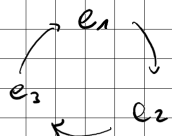
$$\boxed{n=2} \quad \mathbb{R}^2 = \text{span} \{e_1, e_2\}, \quad e_1 \cdot e_2 = 0 \quad \text{and} \quad |e_1| = 1 = |e_2|$$

note: $e_1 \times e_2 \in \mathbb{R}^2 \Rightarrow e_1 \times e_2 = \alpha e_1 + \beta e_2$

$$\begin{aligned}
 & \bullet |e_1 \times e_2| = 1 \Rightarrow \alpha^2 + \beta^2 = 1 \\
 & \bullet e_1 \cdot (e_1 \times e_2) = 0 \Rightarrow \alpha = 0 \\
 & \bullet e_2 \cdot (e_1 \times e_2) = 0 \Rightarrow \beta = 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} & \bullet |e_1 \times e_2| = 1 \Rightarrow \alpha^2 + \beta^2 = 1 \\ & \bullet e_1 \cdot (e_1 \times e_2) = 0 \Rightarrow \alpha = 0 \\ & \bullet e_2 \cdot (e_1 \times e_2) = 0 \Rightarrow \beta = 0 \end{aligned}} \right\} \Rightarrow 0 = 1 \text{ W!}$$

→ there is no cross product in $\mathbb{R}^2$

$$\boxed{n=3} \quad \mathbb{R}^3 = \text{span}\{e_1, e_2, e_3\}$$



$$\begin{aligned}
 e_1 \times e_2 &= e_3 \\
 e_2 \times e_3 &= e_1 \\
 e_3 \times e_1 &= e_2
 \end{aligned}$$

$$\begin{aligned}
 e_1 &\leftrightarrow \hat{i} & \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
 e_2 &\leftrightarrow \hat{j} & \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\
 e_3 &\leftrightarrow \hat{k} & \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}
 \end{aligned}$$

$$\boxed{\text{Lemma}} \quad \bullet (a \times b) \cdot c = -b \cdot (a \times c)$$

$$\bullet a \times (b \times c) - c \times (a \times b) = 2(a \cdot c)b - (b \cdot c)a - (a \cdot b)c$$

$$\begin{aligned}
 \text{proof: } 0 &= (b+c) \cdot (a \times (b+c)) = (b+c) \cdot ((a \times b) + (a \times c)) \\
 &= \underbrace{b \cdot (a \times b)}_{=0} + b \cdot (a \times c) + c \cdot (a \times b) + \underbrace{c \cdot (a \times c)}_{=0}
 \end{aligned}$$

$$\Rightarrow \underbrace{(a \times b) \cdot c}_{=0} = -b \cdot (a \times c) \quad \text{and this is (1).}$$

$$\begin{aligned}
 |a \times (b+d)|^2 &= |a|^2 |b+d|^2 - (a \cdot (b+d))^2 \\
 |(a \times b) + (a \times d)|^2 &= (a \cdot a)(b \cdot b + 2bd + d \cdot d) - (a \cdot b)^2 - 2(a \cdot b)(a \cdot d) - (a \cdot d)^2 \\
 &\Rightarrow |a \times b|^2 + 2(a \times b) \cdot (a \times d) + |a \times d|^2 \quad |a|^2 |b|^2 + 2|a|^2(b \cdot d) + |a|^2 |d|^2
 \end{aligned}$$

$$\begin{aligned}
 \text{So: } & \cancel{|a \times b|^2} + 2(a \times b) \cdot (a \times d) + \cancel{|a \times d|^2} \\
 &= \cancel{|a|^2 |b|^2} + 2|a|^2(b \cdot d) + \cancel{|a|^2 |d|^2} - \cancel{(a \cdot b)^2} - 2(a \cdot b)(a \cdot d) - \cancel{(a \cdot d)^2}
 \end{aligned}$$

$$\Rightarrow (a \times b) \cdot (a \times d) = (a \cdot a)(b \cdot d) - (a \cdot b)(a \cdot d)$$

$$\begin{aligned}
 & \xrightarrow{(1)} -a \times (a \times b) \cdot d = ((a \cdot a)b - (a \cdot b)a) \cdot d \\
 & \xrightarrow{a=a+c} =
 \end{aligned}$$

$$\begin{aligned}
 -(a+c) \times ((a+c) \times b) &= ((a+c) \cdot (a+c))b - ((a+c) \cdot b)(a+c) \\
 &\rightarrow \text{expand and simplify}
 \end{aligned}$$

$$\boxed{\text{Lemma}} \quad \text{If } a, b, c \text{ are mutually orthogonal then}$$

$$a \times (b \times c) = c \times (a \times b)$$

$$\boxed{\text{Definition}} \quad B \subseteq \mathbb{R}^n \text{ is } \underline{A\text{-stable}} \text{ if } \forall a \in A, b \in B \quad a \times b \in B.$$

$$\boxed{\text{proposition}} \quad \text{If a subspace } B \subseteq \mathbb{R}^n \text{ is } A\text{-stable then so is } B^\perp.$$

$$\text{proof: suppose } c \in B^\perp \quad (\forall b \in B \quad b \cdot c = 0)$$

Now take any $a \in A$ and look at

$$b \cdot (a \times c) = -\underbrace{(a \times b)}_{\in B} \cdot c = 0 \Rightarrow a \times c \in B^\perp \quad \square$$

proposition

Suppose $A \subseteq \mathbb{R}^n$ is a subspace closed under $\times$ with an orthonormal basis $\{a_1, \dots, a_k\}$.

Then for any unit vector $b \in A^\perp$, $\{b, b \times a_1, \dots, b \times a_k\} \subseteq A^\perp$ is an orthonormal set

$\Rightarrow$ if $A^\perp \neq 0$, $\dim(A^\perp) \geq \dim(A) + 1$

proof: Take any $a \in A$.

$$a \cdot (a_i \times b) = -b \cdot (\underbrace{a \times a_i}_{0}) = 0 \Rightarrow a_i \times b \in A^\perp$$

Further $a_i \perp b$, both unit vectors $\Rightarrow a_i \times b$ is a unit vector.

$$\begin{aligned} \text{And } (a_i \times b) \cdot (a_j \times b) &= (b \times a_i) \cdot (b \times a_j) \\ &= -(b \times (b \times a_i)) \cdot a_j \\ &= -(\underbrace{(b \cdot a_i)}_{=0} b - \underbrace{(b \cdot b)}_{=1} a_i) \cdot a_j = a_i \cdot a_j = \delta_{ij} \end{aligned}$$

explore $\mathbb{R}^n$, $n > 3$

$$e_1, e_2 \in \mathbb{R}^n, |e_1| = |e_2| = 1, e_1 \cdot e_2 = 0$$

Set $e_3 = e_1 \times e_2$. Check $A = \text{span}\{e_1, e_2, e_3\} \Rightarrow (A, \times) \cong (\mathbb{R}^3, \times)$

Since $n > 3$, $A^\perp \neq 0 \Rightarrow \dim(A^\perp) \geq 4$ and

$$n = \dim(A) + \dim(A^\perp) \geq 3 + 4 = 7$$

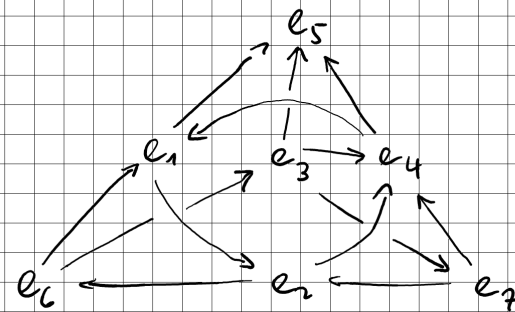
Observations

We have cross products in $\mathbb{R}^1, \mathbb{R}^3, \mathbb{R}^7$

none in $\mathbb{R}^2, \mathbb{R}^4, \mathbb{R}^5, \mathbb{R}^6$. What about $\mathbb{R}^n$, $n > 7$?

Seven Dimensional Cross Product

(related to Octonions)



examples:

$$e_1 \times e_3 = e_7$$

$$e_3 \times e_6 = -e_4$$

$$e_2 \times e_3 = e_5$$

Theorem If $n > 7$ there is no cross product on $\mathbb{R}^n$.

proof: take a copy of $(\mathbb{R}^3, \times)$, $A = \text{span}\{e_1, e_2, e_3\} \subseteq \mathbb{R}^n$.

Find a unit vector $b \in A^\perp$.

Set $B = \text{span}\{e_1, e_2, e_3, b, b \times e_1, b \times e_2, b \times e_3\}$ orthogonal set

Take $c \in B^\perp$, set $d = b \times c$
 $c \times d = b$
 $d \times b = c$

$$\begin{aligned} (e_i \times b) \times (e_j \times c) &= (e_i \times b) \times (e_j \times (d \times b)) \\ &= (e_j \times d) \times ((e_i \times b) \times b) \end{aligned}$$

$$= (e_j \times d) \times (\underbrace{(b \cdot e_i)}_{=0} b - \underbrace{(b \cdot b)}_{=1} e_i)$$

$$= -(e_j \times d) \times e_i$$

$$(e_i \times b) \times (e_j \times c) = d \times (e_i \times e_j)$$

$$\text{similarly: } (e_j \times c) \times (e_i \times b) = d \times (e_i \times e_j)$$

$$\Rightarrow d \times (e_i \times e_j) = 0 \quad \text{but} \quad |d \times (e_i \times e_j)| = 1 \quad \text{W!}$$

Contradiction to $c \in B^\perp$,
 $B^\perp$ shouldn't have existed.
 But the only reason $B^\perp$ existed
 was the fact that $n > 7$.
 So it's impossible for $n > 7$
 and to have a cross product.